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Assignment-2

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Submitted to:-
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Section-A

1) Entity relationship diagram
2) Relational Model
3) Object Oriented Model
4) Identity constraints

Section-B

Ques) In an healthcare system, discuss how data abstraction can simplify the interaction b/w healthcare professionals and patient records.

Ans

Data abstraction in a healthcare system hides complex database details and provide only the necessary information to healthcare professionals. It allows doctors, nurses, & hospitals staff to interact with patient records easily without understanding the internal database structure. This protects sensitive data, Reduces chances of data misuses. Thus, data abstraction helps healthcare focus on patient care instead of technical database complexities.

Ques 2. Explore the potential benefits and challenges associated with implementing data abstraction in this context.

Ans Benefits:-

- Improved data security
- Easier data management
- Data Independence
- Better data consistency
- Efficient system performance.

Challenges of Data abstraction:-

- System complexity
- Development cost
- Performance overhead
- Data synchronization Issues.

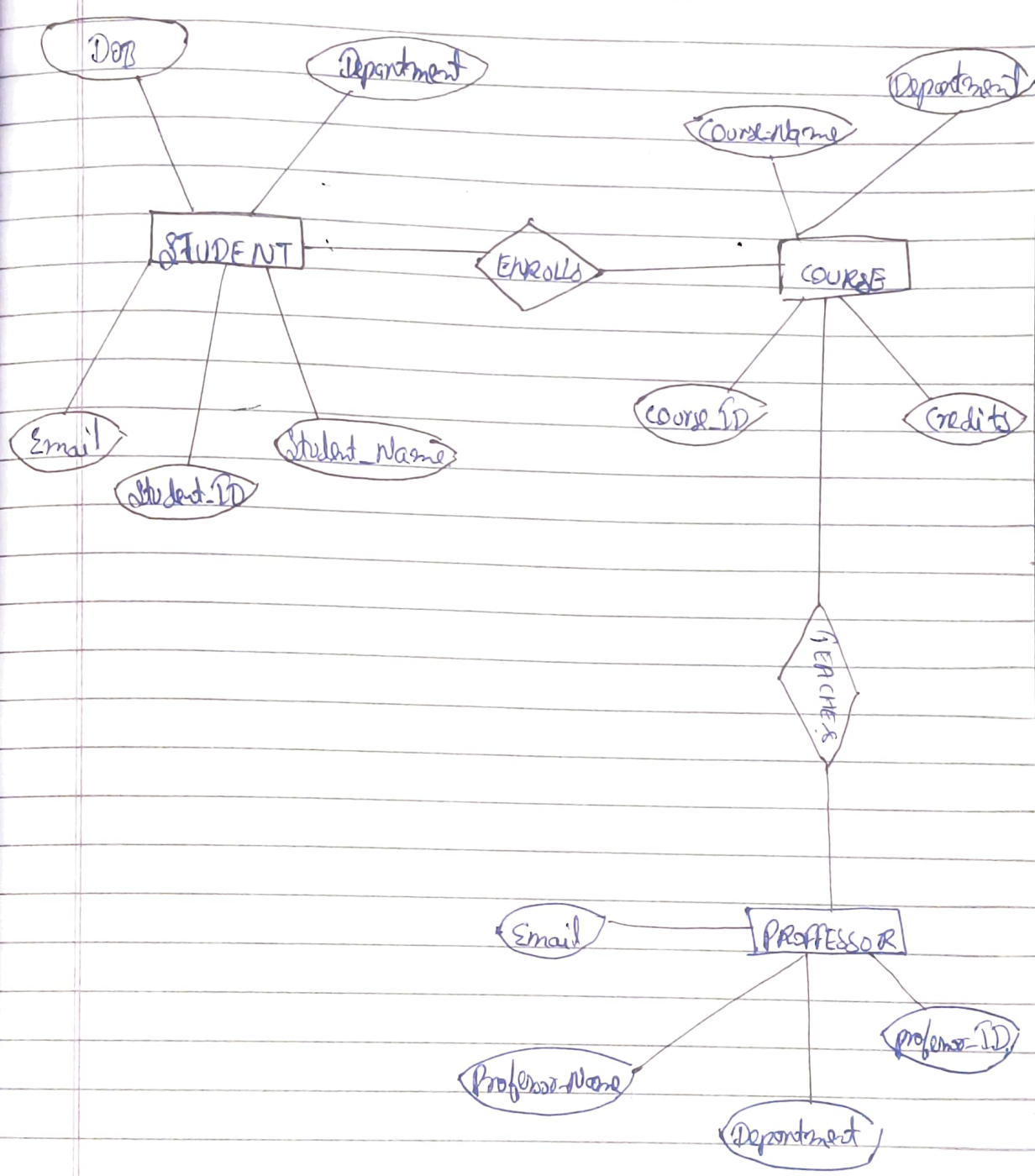
* Section-C

Q3. Design a database for a university. Using the Entity-Relation Diagram.

- Student Entity
  - Student_ID
  - Student_Name
  - Student_DOB
  - Email
  - Department

- Course Entity
  - Course_ID
  - Course_Name
  - Credits
  - Department

- Professor.
  - Professor_ID
  - Professor_Name
  - Email
  - Department.


DOB
Department
STUDENT
ENROLLS
COURSE
Course_Name
Department
Email
Student_ID
Student_Name
Course_ID
Credits
TEACHES
Email
PROFESSOR
Professor_Name
Professor_ID
Department